

Yinjie Yu

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EDUCATION

Princeton University

Ph.D. candidate, Economics - Finance and Macroeconomics; GPA: 3.89/4.00

New Jersey, US

Aug 2021 – Feb 2027 (expected)

- **Teaching:** TA (Preceptor) for: Fin 521 Fixed Income, Options, and Derivatives (Spring 2024–2026); Fin 515 Portfolio Theory and Asset Management (Fall 2024–2026); Econ 101 Macroeconomics (Spring 2024).

Peking University

B.A. in Economics (Minor in Mathematics and Applied Mathematics); GPA: 3.82/4.00

Beijing, China

Sep 2017 – Jul 2021

- **Awards:** 2018 Leo KoGuan Scholarship (Top 5%); Merit Student of PKU (2017, 2018); 2021 Excellent Honor Thesis of Beijing.

WORKING EXPERIENCE

Sciencetech Research LLC

Quant Researcher Intern

City, State

Month 2026 – Present

- High-frequency alpha research on level 3 Nasdaq order book. Constructed the inward order replacement imbalance alpha.
- Strategy detection: trades categorized by the Hawkes model and TWAP intensity based on max-pooled spectrum analysis.
- Alpha based on the fill rate imbalance of counterfactual orders (simulation based) with spread-dependent aggressiveness and patience, triggered by various trade and quote patterns (pull/size-up).

IMC Trading

Quant Researcher Intern

Chicago, Illinois

Jun 2025 – Aug 2025

- Developed a cross-hedging strategy for index futures pairs (NQ-ES, RTY-ES) under market-maker delta constraints.
- Constructed features (intra-second frequency, processed by SQL) from price (z-score, beta), volume, and volatility data.
- Performed spread prediction and feature selection using Lasso regression and XGBoost to generate trading signals.
- Simulated strategy impact on MM position. Backtested and optimized over hedging regimes, features, and parameterizations.

The International Monetary Fund

Fund Intern, European Department

Washington, DC

Jun 2024 – Aug 2024

- Developed a framework to assess fiscal revenue risk and debt sustainability under de-globalization and capital-outflow.
- Estimated a panel VAR model of FDI for small open economies and simulated fiscal revenue, output, and debt dynamics.
- Measured firm-concentration risk and evaluated Ireland's fiscal sustainability under different shock and policy scenarios.
- Constructed an optimal fiscal rule facing a stabilization-sustainability trade-off (Minimax Robustness Control).

GF Securities Co., Ltd.

Macro Strategy Analyst Intern

Shanghai, China

Jun 2020 – Sep 2020

- Wrote analytical reports on monetary policy, construction-sector spillovers (I/O matrix analysis), real estate financing regulation, corporate bond default risk, and global value chain shifts in autos and consumer electronics.

RESEARCH EXPERIENCE

Bendheim Center for Finance, Princeton University

Research Assistant to Prof. [Moritz Lenel](#)

Princeton, NJ

Oct 2021 – Present

- **Monetary Policy, Segmentation, and the Term Structure:** Solved bond-pricing PDEs under arbitrageurs' time-varying risk aversion using deep learning: trained a neural network to approximate the bond yield surface by minimizing PDE residuals via automatic differentiation, with dynamic sampling weighted by local residual loss.
- **Monetary Policy, Redistribution, and Risk Premia:** Globally solved a heterogeneous-agent DSGE model with portfolio choice in Julia (Smolyak sparse grids, Gauss-Hermite quadrature, Brent's method, value function iteration).

WORKING PAPER/PROJECT

- **Predicting Stock Returns with Financial News:** We fine-tune FinBERT's attention layers with an MLP trading head replacing its sentiment head; the predicted trading propensity forecasts next-day intraday returns for Nasdaq-listed firms.
- **A Model of Optimal QE with Segmented Bond Market and Liquidity Regulations:** I propose a theory of the optimal design of asset-purchase program (QE) and exit policy (QT) in response to financial shocks with a banking sector facing macro-prudential liquidity regulations, which make reserves a key factor of financial intermediation efficiency and leverage capacity.
- **Self-Fulfilling Inflation Risk Premia under the Fiscal Theory:** A New Keynesian bond-pricing model admits a Fundamental and a self-validating Sunspot Equilibrium; QE revalues debt via this risk-price channel, dominating the term-premium channel.

ADDITIONAL INFORMATION

- **Programming Skills:** Python (TensorFlow, PyTorch), SQL, Julia, MATLAB, Stata, C++.
- **Language:** English: TOEFL 110 (Speaking 25); GRE V157 Q170 AW4.5. Chinese Mandarin: Native. Japanese: Intermediate.
- **Math Awards:** H Prize in 2020 MCM; First Prize in Chinese Math Olympiad (2015, 2016); Gold Medal in 2016 SE China MO.
- **Interests:** Basketball, Poker, Photography, Piano (Level 10, choir accompanist), Civ6.